GEOSPATIAL AI & ENVIRONMENTAL MACHINE LEARNING

Urban Heat Island

Predictive modeling of Land Surface Temperature (LST) using satellite telemetry and spatial AI.

github.com/tmthangaraj1980-AIML/Urban_Heat_Island

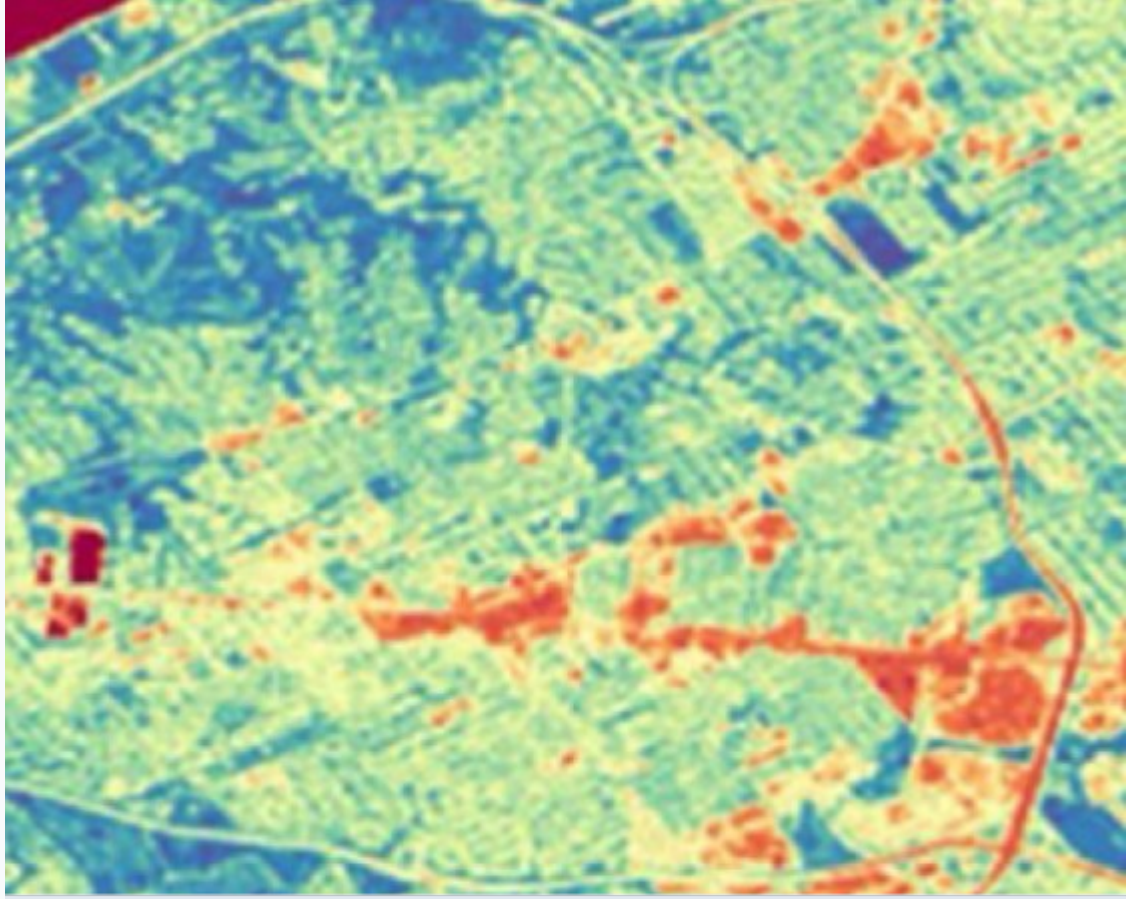
THE PROBLEM

Thermal Inequity

Microclimate Extremes: Dense concrete traps heat, raising localized temperatures.

Resource Strain: Exponential spikes in urban AC energy consumption.

Invisible Vulnerability: City planners lack high-resolution thermal mapping.



Solution Objectives

Earth Observation

Process Landsat/Sentinel multispectral imagery streams.

Index Extraction

Calculate NDVI (Vegetation) and NDBI (Built-up) ratios.

Spatial Prediction

Generate high-resolution predictive thermal heatmaps.

Geospatial Tech Stack



Python



Google Earth Engine



GeoPandas



Rasterio

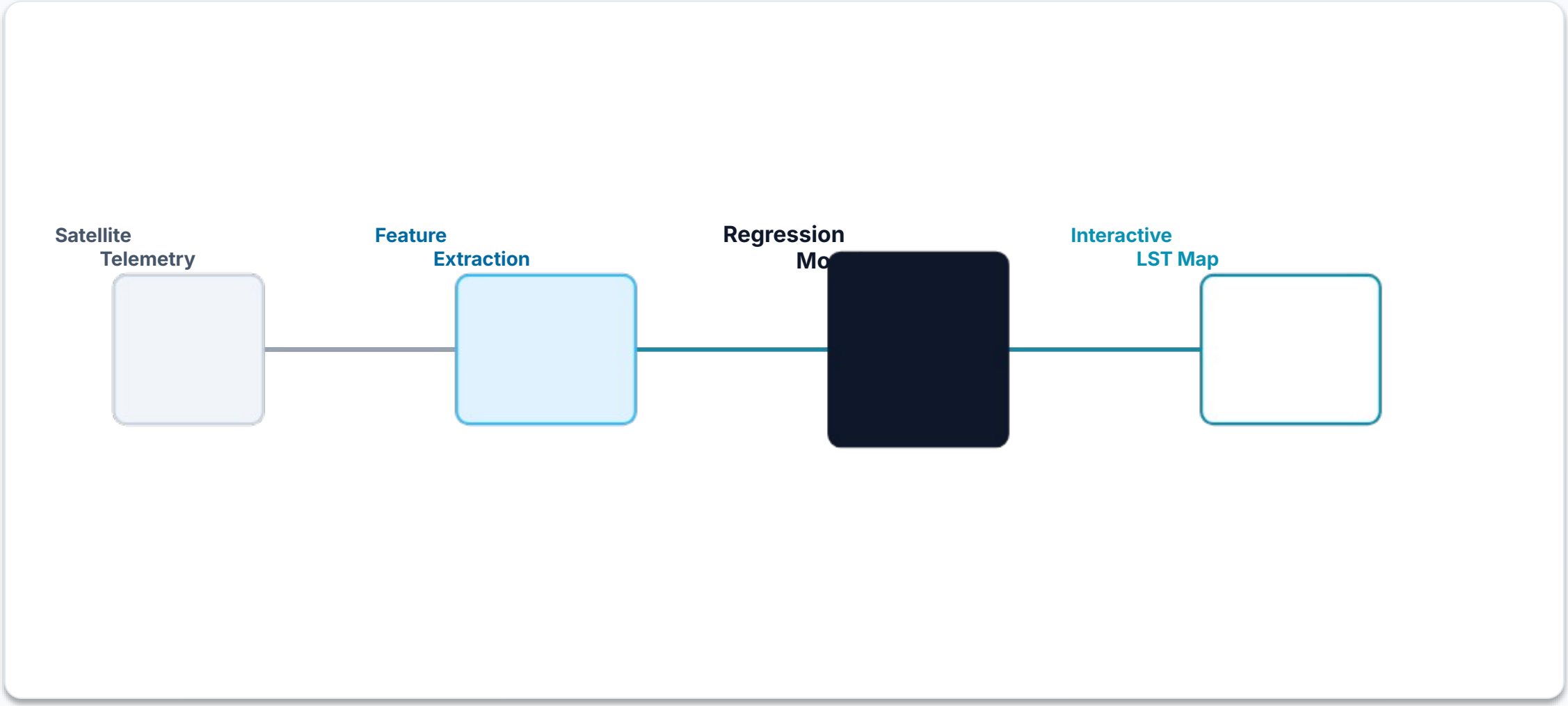


Scikit-Learn



Streamlit

Processing Architecture



Geospatial Pipeline

1. Ingestion

Fetch Landsat 8 TIRS/OLI bands.

2. Alignment

Cloud masking and CRS projection.

3. Engineering

Compute NDVI, NDBI, and Albedo.

4. Inference

Predict spatial Land Surface Temp.

Data Sources & Key Vectors

Inputs

- Landsat 8 Operational Land Imager.
- Thermal Infrared Sensor (TIRS) Band 10.
- OpenStreetMap Building Footprints.

Derived Features

- NDVI: Normalized Difference Vegetation Index.
- NDBI: Normalized Difference Built-up Index.
- Albedo: Surface solar reflectance ratio.

Predictive Modeling

Regression Engine

Algorithm: Random Forest / XGBoost

Maps non-linear relationships between spectral indices (NDVI/NDBI) and ground truth temperature arrays.

Spatial Upsampling

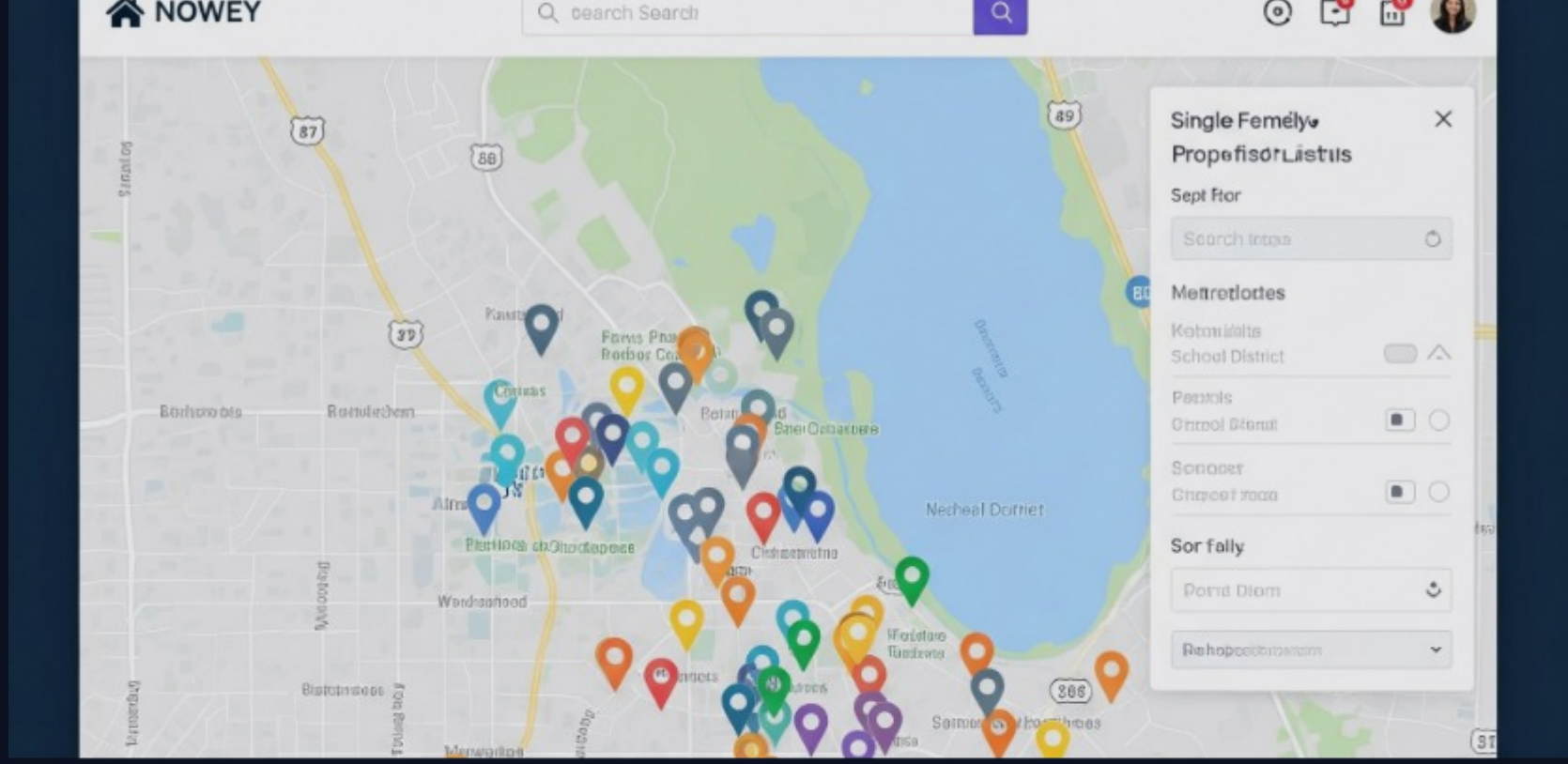
Architecture: Convolutional Filter

Enhances 100m thermal band resolution using 30m multispectral bands for granular block-level mapping.

Interactive Dashboard

Folium and Streamlit integration projecting LST inference layers directly over interactive city basemaps.

Data Layer: Active


NOWEY
Single Family
Property Details
Sept filter
Search maps
Metropolitan
School District
Parcels
Sold
Sort by
Pond Diem
Rehoboth Beach
Bethany Beach
Fenwick Island
Wenonah
Woodland
Millville
Dagsboro
Frankford
Selbyville
Ocean View
Millsboro
Georgetown
Lewes
Rehoboth
Milton
Harbeson
Long Neck
Indian River
Roxana
Angola
Bayside
Dewey Beach
Clarksville
Fenwick
Bethany
Ocean
Delaware
Bay
Indian
River
Bay
Rehoboth
Beach
Bethany
Beach
Fenwick
Island
Lewes
Beach
Long
Neck
Millsboro
Pond
Georgetown
Milton
Harbeson
Roxana
Angola
Bayside
Dewey
Beach
Clarksville
Ocean
View
Frankford
Selbyville
Millville
Dagsboro
Wenonah
Woodland
Bethany
Beach
Ocean
Delaware
Bay
Indian
River
Bay
Rehoboth

Results & Constraints

Model Performance

1.2°C	0.89
Root Mean Square Error	R-Squared Score

Technical Limitations

- **Optical occlusion from heavy cloud cover.**
- **Low temporal resolution (16-day passes).**
- **Missing micro-wind fluid dynamic factors.**

Technical Q&A

1. How is LST derived physically?

Converting Top of Atmosphere (TOA) spectral radiance to brightness temperature, then correcting for surface emissivity.

2. Role of Google Earth Engine?

Cloud-based API allowing petabyte-scale satellite raster computation without local downloads.

3. Handling CRS (Coordinate Reference System) mismatches?

Reprojecting all layers to a unified UTM zone using GeoPandas/Rasterio.

4. Why use NDVI as a feature?

Vegetation transpires water, actively cooling surfaces; strong negative correlation with LST.

5. Explain Spatial Autocorrelation?

Tobler's First Law: near things are more related. Addressed by spatial cross-validation.

Mapping the Climate Micro-Front.

Empowering urban planners with high-fidelity thermal intelligence to deploy targeted green infrastructure.

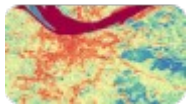


/Urban_Heat_Island



tmthangaraj1980

Image Sources



https://earthdata.nasa.gov/s3fs-public/styles/hero_background_745/public/2025-09/ARSET-UrbanHeat-2022-hero.png?VersionId=MYHJ.pzw4CMKCitn1Vibbn5eWhz4nhIU&itok=08ffTUq4

Source: www.earthdata.nasa.gov



https://www.aeologic.com/images/interactive-web-map-for-real-estate-listings.png

Source: www.aeologic.com